

SAPTHAGIRI COLLEGE OF ENGINEERING, BENGALURU-560 057.
(Affiliated to VTU, Belagavi & Approved by AICTE, New Delhi)
DEPARTMENT OF MATHEMATICS
QUESTION BANK (I SEM)
ACADEMIC YEAR: 2023-24 (ODD SEMESTER)



MODULE-2: DIFFERENTIAL CALCULUS-2

Sub: MATHEMATICS-I

Sub Code: BMATE/C/S/M101

1. Obtain the Maclaurin's Series Expansion of the function $\log_e(1+x)$ up to 3^{rd} degree term.
2. Using Maclaurin's expansion Prove that $\sqrt{1+\sin 2x} = 1+x - \frac{x^2}{2} - \frac{x^3}{6} + \frac{x^4}{24}$
3. Obtain the Maclaurin's Series Expansion of the function $\log_e \sec x$ up to 3^{rd} degree term.
4. Evaluate : $\lim_{x \rightarrow 0} \left(\frac{\tan x}{x} \right)^{1/x^2}$
5. Evaluate : $\lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right)^{1/x^2}$
6. Evaluate : $\lim_{x \rightarrow 0} \left[\frac{a^x + b^x + c^x + d^x}{4} \right]^{1/x}$
7. If $u = f(x-y, y-z, z-x)$ then Prove that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0$.
8. If $u = f\left(\frac{x}{y}, \frac{y}{z}, \frac{z}{x}\right)$ Prove that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = 0$.
9. If $u = f(2x-3y, 3y-4z, 4z-2x)$. then Prove that $\left[\frac{1}{2}\right] \frac{\partial u}{\partial x} + \left[\frac{1}{3}\right] \frac{\partial u}{\partial y} + \left[\frac{1}{4}\right] \frac{\partial u}{\partial z} = 0$
10. If $r^2 = (x-a)^2 + (y-b)^2 + (z-c)^2$ show that $\frac{\partial^2 r}{\partial x^2} + \frac{\partial^2 r}{\partial y^2} + \frac{\partial^2 r}{\partial z^2} = \frac{2}{r}$
11. Find the extreme values of the function $f(x, y) = x^3 + y^3 - 3x - 12y + 20$.
12. Examine the function $f(x, y) = 2(x^2 - y^2) - x^4 + y^4$ for its extreme values.
13. Examine the function $f(x, y) = 2 + 2x + 2y - x^2 - y^3$ for its extreme values.
14. If $u = \frac{yz}{x}, v = \frac{xz}{y}, w = \frac{xy}{z}$ Find $\frac{\partial(u, v, w)}{\partial(x, y, z)}$.

15. If $x + y + z = u$, $y + z = v$, $z = uvw$ Find $J\left(\frac{x,y,z}{u,v,w}\right)$.

16. If $x = r\sin\theta\cos\phi$, $y = r\sin\theta\sin\phi$, $z = r\cos\theta$ Find $J\left(\frac{x,y,z}{r,\theta,\phi}\right)$.